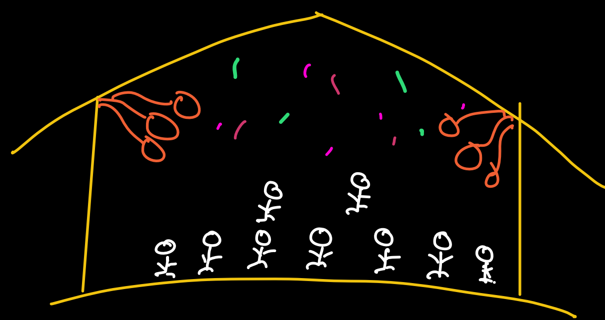
Q Ways to party

Given N people inside a party hall.

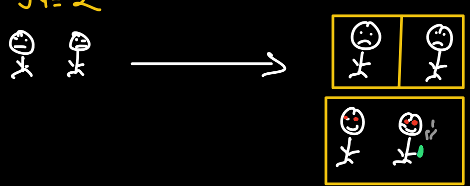
1. Being alone
2. Getting paired with someone

Return no of ways in which party can be enjoyed.

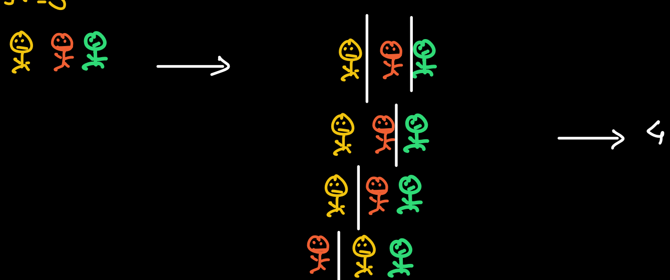


$N = 1$

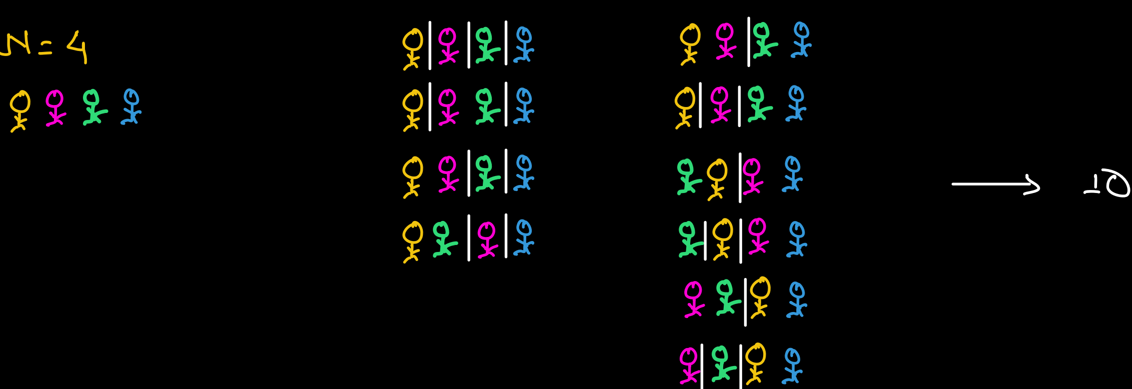
$N = 2$

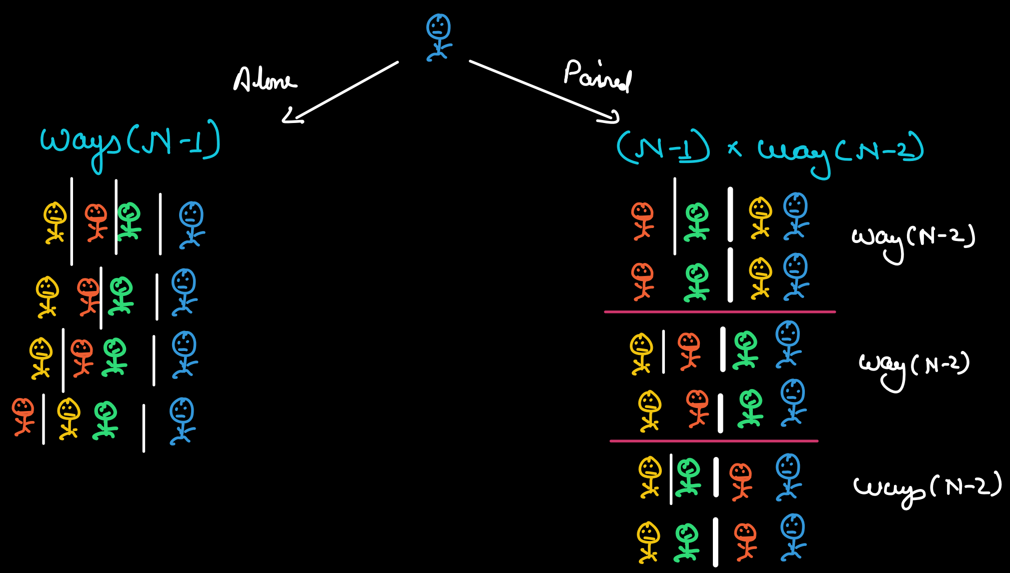


$N = 3$



$N = 4$





$$\text{Way}(N) = \text{Ways}(N-1) + (N-1) \times \text{Way}(N-2)$$

HW : Draw for N=5

Base Cases

$N = 1 \longrightarrow 1$

$N = 2 \longrightarrow 2$

Q. Ways to decode

Vishnu $\longrightarrow$ Giridharan

A $\longrightarrow$ 1

Prakash

$A \to 1$
$B \to 2$
$C \to 3$
$\vdots$
$Z \to 26$

1	$\longrightarrow$	1
12	$\longrightarrow$	2
121	$\longrightarrow$	3
26	$\longrightarrow$	2

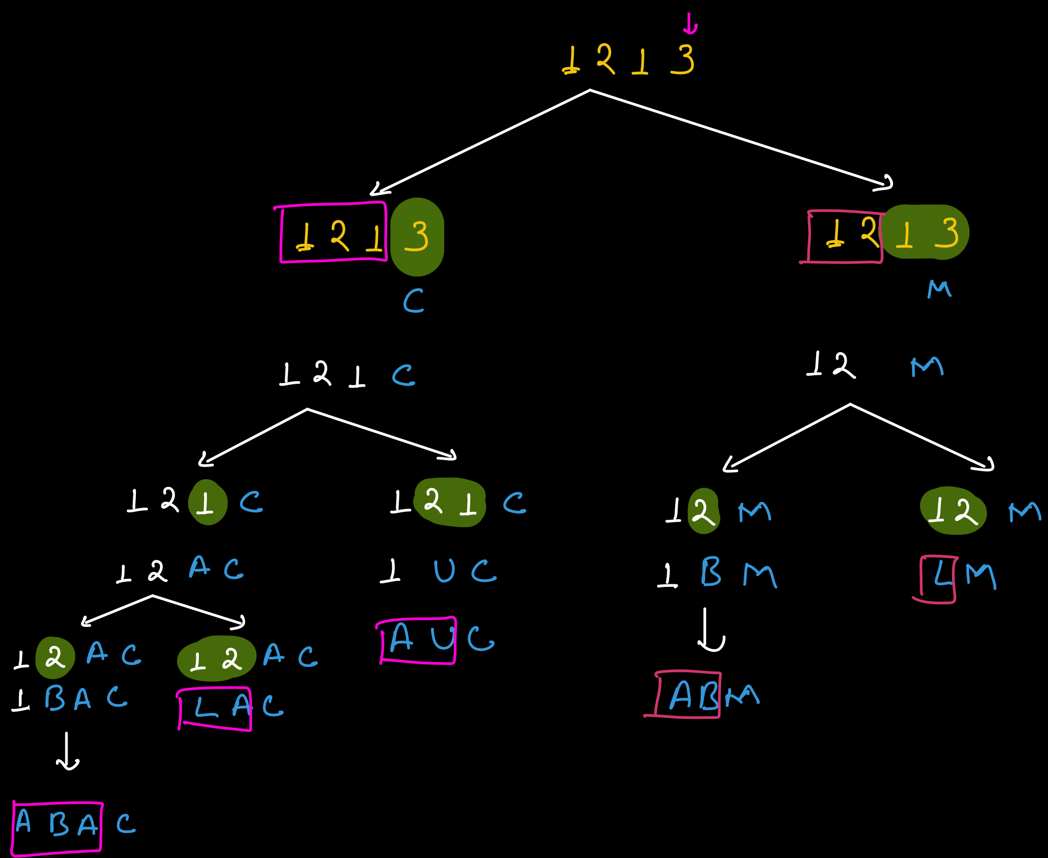
1

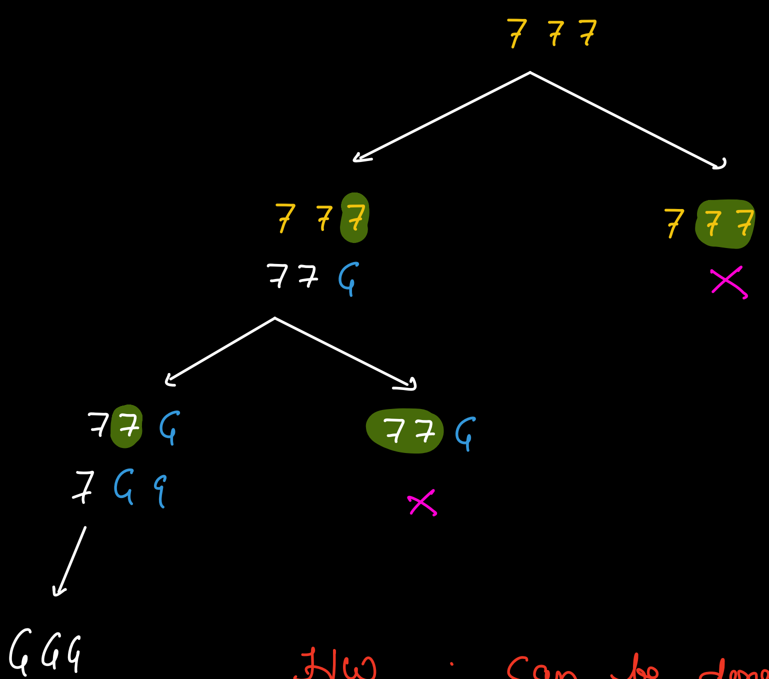
12 → AB
 → L

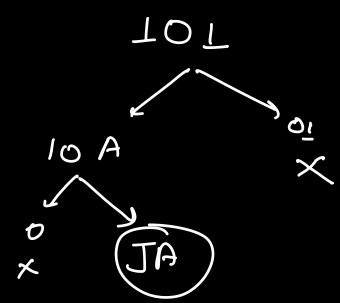
121 → ABA
 → LA
 → AU

26 → BF
 → Z

Choice : Consider the i^{th} digits as a single charact
OR
Pair it with $(i-1)^{th}$ digit (if possible)







HW : Can be done from left → right ?

ways(i) → No of ways of decoding the string from index 0 to i.

$$way(i) = \underset{A[i]>0}{ways(i-1)} + \underset{A[i-1]\times 10 + A[i] <= 26 \;\&\&\; A[i-1]>0}{ways(i-2)}$$

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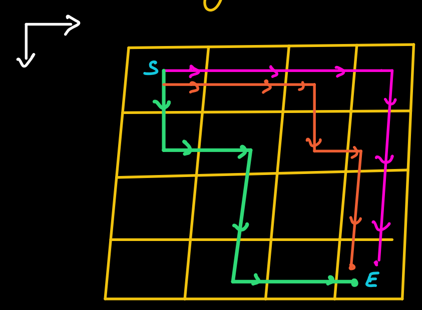
*

- → Write recursion
- → Do memoization
- → Do dry run to observe how array is getting filled
- → Write iterative code

Q. Given a 2D matrix of size $N \times M$.
Count the no of unique paths to reach $(N-1, M-1)$ from $(0, 0)$.
From any cell we can either move right or down.



$$(0,0) \rightarrow (0,1)$$
$$(0,0) \downarrow (1,0)$$

Paths ((0,0) to (N-1, M-1)) = Paths ((0,1) to (N-1, M-1)) + Paths ((1,0) to (N-1, M-1))

Paths (i, j) ⟶ No. of path from i, j to (N-1, M-1)

paths (i, j) = path (i, j+1) + path (i+1, j)

N x M

N > M

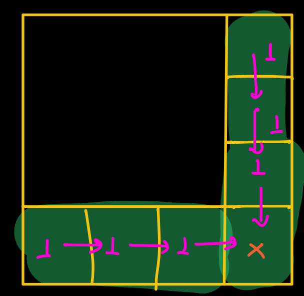
path (i, j)
- path (i, j+1)
  - Path (i, j+2)
    - ⋮
    - Path (i, M-1)
  - Path (i+1, j+1)
    - ⋮
- Path (i+1, j)
  - Path (i+1, j+1)
  - Path (i+2, j)
    - ⋮
    - Path (N-1, j)

Path(N-1, M-1)

Paths(i,j) $\longrightarrow$ No. of paths from i,j to (N-1, M-1)

$$paths(i,j) = paths(i, j+1) + paths(i+1, j)$$

if(i == N-1 || j == M-1)
DP[i][j] = 1;



Start from base case

HW: Optimize SC.

126	56	21	6	1
70	35	15	5	1
35	20	10	4	1
15	10	6	3	1
5	4	3	2	1
1	1	1	1	E